JEE (Main) QUESTION PAPER 2025

22nd Jan (Shift 1)

Time : 3 hours **Total Marks : 300**

General Instructions :

1. *There are three subjects in the question paper consisting of Physics (Q. no. 1 to 25), Chemistry (Q. no. 26 to 50) and Mathematics (Q. no. 51 to 75).*
2. *Each subject is divided into two sections. Section A consists of 20 multiple choice questions & Section B consists of 5 numerical value type questions. In Section B, candidates have to attempt any five questions out of 5.*
3. *There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice for Section A questions and zero mark will be awarded for not attempted question.*
4. *For Section B questions, 4 marks will be awarded for correct answer and zero for unattempted and incorrect answer.*
5. *Any textual, printed or written material, mobile phones, calculator, etc. is not allowed for the students appearing for the test.*
6. *All calculations / written work should be done in the rough sheet is provided with Question Paper.*

Physics

Section A

Q. 1. Given below are two statements:

Statement I: In a vernier calliper, one vernier scale division is always smaller than one main scale division.

Statement II: The vernier constant is given by one main scale division multiplied by the number of vernier scale divisions.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are false.

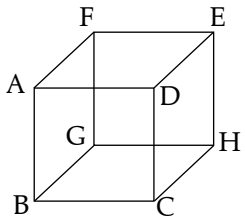
(2) Statement I is true but Statement II is false.

(3) Both Statement I and Statement II are true.

(4) Statement I is false but Statement II is true.

Q. 2. A line charge of length $\frac{a}{2}$ is kept at the centre of an edge 'BC' of a cube ABCDEFGH having edge length 'a' as shown in the figure. If the density of line is λC per unit length, then the total electric flux through all the faces of the cube will be ____.

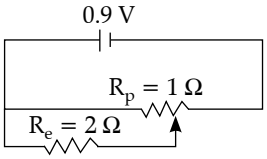
(Take ε_0 as the free space permittivity.)



(1) $\frac{\lambda a}{8\varepsilon_0}$ **(2)** $\frac{\lambda a}{16\varepsilon_0}$

(3) $\frac{\lambda a}{2\varepsilon_0}$ **(4)** $\frac{\lambda a}{4\varepsilon_0}$

Q. 3.



Sliding contact of a potentiometer is in the middle of the potentiometer wire having resistance $R_P = 1\ \Omega$ as shown in the figure. An external resistance of $R_e = 2\ \Omega$ is connected via the sliding contact, The current i is:

(1) 0.3 A **(2)** 1.35 A **(3)** 1.0 A **(4)** 0.9 A

Q. 4. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): If Young's double slit experiment is performed in an optically denser medium than air, then the consecutive fringes come closer.

Reason (R): The speed of light reduces in an optically denser medium than air while its frequency does not change.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) (A) is false but (R) is true.

(3) Both (A) and (R) are true but (R) is not the correct explanation for (A).

(4) (A) is true but (R) is false.

Q. 5. Two spherical bodies of same materials having radii 0.2 m and 0.8 m are placed in the same atmosphere. The temperature of the smaller body is 800 K and temperature of the bigger body is 400 K. If the energy radiated from the smaller body is E, the energy radiated from the bigger body is_____ (assume effect of the surrounding to be negligible).

(1) 256 E **(2)** E **(3)** 64 E **(4)** 16 E

Q. 6. An amount of ice of mass 10^{-3} kg and temperature –10°C is transformed to vapour of temperature 110° by applying heat. The total amount of work required for this conversion is:

(Take the specific heat of ice = 2100 $Jkg^{-1}K^{-1}$, the specific heat of water = 4180 $Jkg^{-1}K^{-1}$, specific the heat of steam = 1920 $Jkg^{-1}K^{-1}$, the latent heat of ice = 3.35×10^5 Jkg^{-1} and the latent heat of steam = 2.25×10^6 Jkg^{-1}).

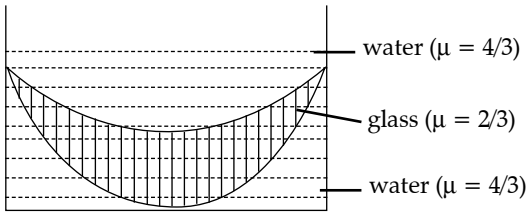
(1) 3022 J **(2)** 3043 J

(3) 3003 J **(4)** 3024 J

Q. 7. An electron in the ground state of the hydrogen atom has the orbital radius of 5.3×10^{-11} m while that for the electron in the third excited state is 8.48×10^{-10} m. The ratio of the de Broglie wavelengths of electron in the excited state to that in ground state is:

(1) 4 **(2)** 9 **(3)** 3 **(4)** 16

Q. 8. In the diagram given below, three lenses are formed. Considering the negligible thickness of each of them as compared to $[R_1]$ and $[R_2]$, i.e., the radii of curvature for the upper and lower surfaces of the glass lens, the power of the combination is:



(1) $-\frac{1}{6}\left(\frac{1}{|R_1|}+\frac{1}{|R_2|}\right)$ **(2)** $-\frac{1}{6}\left(\frac{1}{|R_1|}-\frac{1}{|R_2|}\right)$

(3) $\frac{1}{6}\left(\frac{1}{|R_1|}+\frac{1}{|R_2|}\right)$ **(4)** $\frac{1}{6}\left(\frac{1}{|R_1|}-\frac{1}{|R_2|}\right)$

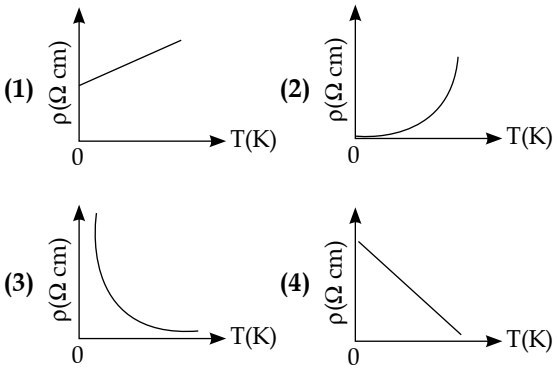
Q. 9. An electron is made to enter symmetrically between two parallel and equal but oppositely charged metal plates, each 10 cm in length. The electron emerges out of the field region with a horizontal component of velocity 10^6 m/s. If the magnitude of the electricity field between the plates is 9.1 V/cm, then the vertical component of the velocity of the electron is:

(Mass of electron = 9.1×10^{-31} kg and charge of electron = 1.6×10^{-19} C.)

(1) 1×10^6 m/s **(2)** 0

(3) 16×10^6 m/s **(4)** 16×10^4 m/s

Q. 10. Which of the following resistivity (ρ) v/s temperature (T) curves is most suitable to be used in wire-bound standard resistors?

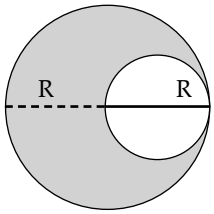


Q. 11. A closed organ and an open organ tube filled with two different gases have the same bulk modulus but different densities ρ_1 and ρ_2, respectively. The frequency of the 9th harmonic of the closed tube is identical to the 4th harmonic of the open tube. If the length of the closed tube is 10 cm and the density ratio of the gases is $\rho_1 : \rho_2 = 1 : 16$, then the length of the open tube is:

(1) $\frac{20}{7}$cm **(2)** $\frac{15}{7}$cm

(3) $\frac{20}{9}$cm **(4)** $\frac{15}{9}$cm

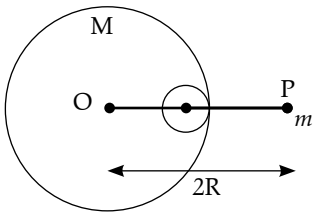
Q. 12. A uniform circular disc of radius 'R' and mass 'M' is rotating about an axis perpendicular to its plane and passing through its centre. A small circular part of radius $\frac{R}{2}$ is removed from the original disc as shown in the figure. Find the moment of inertia of the remaining part of the original disc about the axis as given above.



(1) $\frac{7}{32}MR^2$ **(2)** $\frac{9}{32}MR^2$

(3) $\frac{17}{32}MR^2$ **(4)** $\frac{13}{32}MR^2$

Q. 13. A small point of mass m is placed at a distance of 2R from the centre 'O' of a big uniform solid sphere of mass M and radius R. The gravitational force on 'm' due to M is F_1. A spherical part of radius R/3 is removed from the big sphere as shown in the figure and the gravitational force on m due to the remaining part of M is found to be F_2. The value of ratio $F_1 : F_2$ is:



(1) 16 : 9 **(2)** 11 : 10

(3) 12 : 11 **(4)** 12 : 9

Q. 14. The work functions of cesium (Cs) and lithium (Li) metals are 1.9 eV and 2.5 eV, respectively. If we incident a light of wavelength 550 nm on these two metal surfaces, then the photoelectric effect is possible in the case of:

(1) Li only **(2)** Cs only

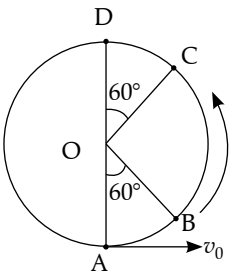
(3) Neither Cs nor Li **(4)** Both Cs and Li

Q. 15. If B is a magnetic field and μ_0 is the permeability of free space, then the dimensions of (B/μ_0) is/are:

(1) $[MT^{-2}A^{-1}]^2$ **(2)** $[L^{-1}A]^2$

(3) $[LT^{-2}A^{-1}]^2$ **(4)** $[ML^2T^{-2}A^{-1}]^2$

Q. 16. A bob of mass m is suspended at a point 'O' by a light string of length l and is left to perform vertical motion (circular) as shown in the figure. Initially, by applying horizontal velocity v_0 at the point 'A,' the string becomes slack when the bob reaches point 'D.' The ratio of the kinetic energy of the bob at points 'B' and 'C' is _____.



(1) 2 **(2)** 1 **(3)** 4 **(4)** 3

Q. 17. Given below are two statements:

Statement I: The equivalent emf of two non-ideal batteries connected in parallel is smaller than either of the two emfs.

Statement II: The equivalent internal resistance of two non-ideal batteries connected in parallel is smaller than the internal resistance of either of the two batteries.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false.

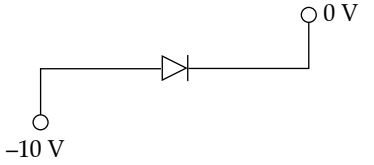
(2) Both Statement I and Statement II are false.

(3) Both Statement I and Statement II are true.

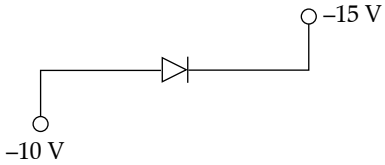
(4) Statement I is false but Statement II is true.

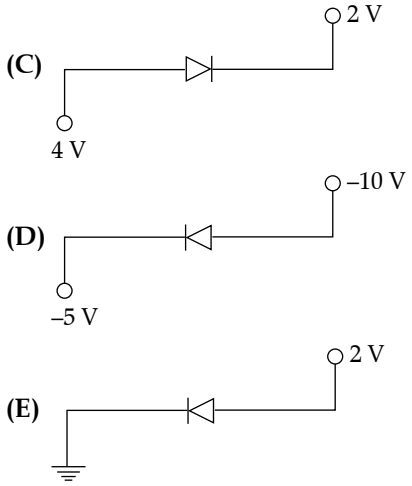
Q. 18. Which of the following circuits represent a forward-biased diode ?

(A)



(B)





Choose the correct answer from the options given below:

(1) (B), (D) and (E) only.

(2) (A) and (D) only.

(3) (B), (C) and (E) only.

(4) (C) and (E) only.

Q. 19. A parallel plate capacitor of capacitance 40 μF is connected to a 100 V power supply. Now, the intermediate space between the plates is filled with a dielectric material of dielectric constant K = 2. Due to the introduction of dielectric material, the extra charge and the change in the electrostatic energy in the capacitor, respectively, are:

(1) 2 mC and 0.2 J **(2)** 8 mC and 2.0 J

(3) 4 mC and 0.2 J **(4)** 2 mC and 0.4 J

Q. 20. Given is a thin convex lens of glass (refractive index μ) and each side has a radius of curvature R. One side is polished for complete reflection. At what distance from the lens, an object should be placed on the optic axis so that an image gets formed on the object itself?

(1) R/μ **(2)** R/(2μ–3)

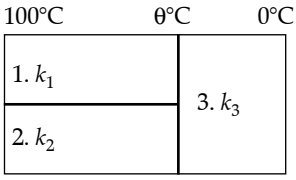
(3) μR **(4)** R/(2μ–1)

Section B

Q. 21. Two soap bubbles of radius 2 cm and 4 cm, respectively, are in contact with each other. The radius of curvature of the common surface, (in cm) is _____ .

Q. 22. A driver sitting inside a parked car is watching vehicles approaching from behind with the help of his side view mirror, which is a convex mirror with radius of curvature R = 2 m. Another car approaches him from behind with a uniform speed of 90 km/h. When the car is at a distance of 24 m from him, the magnitude of the acceleration of the image in the side view mirror is '*a*'. The value of 100*a* is _____ m/s^2.

Q. 23. Three conductors of same length having thermal conductivity k_1, k_2 and k_3 are connected as shown in the figure below.



The of cross-section of the 1st and 2nd conductor is the same and for the 3rd conductor, it is double than that of the 1st conductor, The temperatures are given in the figure above. In steady state condition, the value of θ is _____ °C.

(Given : $k_1 = 60\ Js^{-1}m^{-1}K^{-1}$, $k_2 = 120\ Js^{-1}m^{-1}K^{-1}$, $k_3 = 135\ Js^{-1}m^{-1}K^{-1}$).

Q. 24. The position vectors of two 1 kg particles, (A) and (B), are given by $\vec{r}_A = \left(\alpha_1 t^2\hat{i} + \alpha_2 t\hat{j} + \alpha_3 t\hat{k}\right)m$ and $\vec{r}_B = \left(\beta_1 t\hat{i} + \beta_2 t^2\hat{j} + \beta_3 t\hat{k}\right)m$ respectively ; (α_1 = 1 m/s^2, $\alpha_2 = 3n$ m/s, α_3 = 2 m/s, β_1 = 2 m/s, β_2 = –1 m/s^2, and $\beta_3 = 4p$ m/s), where, *t* is time and *n* and *p* are constants. At $t = 1$ s, $|\vec{V}_A| = |\vec{V}_B|$ and velocities $\vec{V}_A$ and $\vec{V}_B$ of the particles are orthogonal to each other. At $t = 1$ s, the magnitude of angular momentum of particle (A) with respect to the position of particle (B) is $\sqrt{L}$ kgm^2s^{-1}. The value of L is _____.

Q. 25. A particle is horizontally projected at an angle of 30° at a speed of 60 m/s. The height traversed by the particle in the first second is h_0 and the height traversed in the last second, before it reaches the maximum height, is h_1. The ratio $h_0 : h_1$ is _____.

[Take, g = 10 m/s^2]

Chemistry

Section A

Q. 26. A solution of aluminium chloride is electrolysed for 30 min using a current of 2A. The amount of the aluminium deposited at the cathode is ___ .

(Given: Molar mass of aluminium and chlorine is 27 g mol^{-1} and 35.5 g mol^{-1}, respectively, Faraday constant = 96,500 C mol^{-1})

(1) 1.660 g **(2)** 1.007 g

(3) 0.336 g **(4)** 0.441 g

Q. 27 Which of the following statement is not true for radioactive decay?

(1) Amount of radioactive substance remained after three half lives is $\frac{1}{8}$th of original amount.

(2) Decay constant does not depend upon temperature.

(3) Decay constant increases with increase in temperature.

(4) Half-life is ln2 times of $\frac{1}{\text{rate constant}}$.

Q. 28. How many different stereoisomers are possible for the given molecule?

$CH_3–\underset{\displaystyle OH}{\underset{|}{CH}}–CH=CH–CH_3$

(1) 3 **(2)** 1 **(3)** 2 **(4)** 4

Q. 29. Which of the following electronegativity order is incorrect?

(1) Al < Mg < B < N

(2) Al < Si < C < N

(3) Mg < Be < B < N

(4) S < Cl < O < F

Q. 30. Lanthanoid ions with $4f^7$ configuration are:

(A) Eu^{2+} (B) Gd^{3+} (C) Eu^{3+} (D) Tb^{3+} (E) Sm^{2+}

Choose the correct answer from the options given below:

(1) (A) and (B) only

(2) (A) and (D) only

(3) (B) and (E) only

(4) (B) and (C) only

Q. 31. Match List-I with List-II:

List-I	List-II
(A) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$	**(I)** Ionisation enthalpy
(B) B < C < O < N	**(II)** Metallic character
(C) B < Al < Mg < K	**(III)** Electronegativity
(D) Si < P < S < Cl	**(IV)** Ionic radii

Choose the correct answer from the options given below:

(1) A-IV, B-I, C-III, D-II

(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-IV, C-II, D-I

Q. 32. Which of the following acids is a vitamin?

(1) Adipic acid **(2)** Aspartic acid

(3) Ascorbic acid **(4)** Saccharic acid

Q. 33. A liquid when kept inside a thermally insulated closed vessel at 25° C was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters?

(1) $\Delta U > 0, q = 0, w > 0$

(2) $\Delta U = 0, q = 0, w = 0$

(3) $\Delta U < 0, q = 0, w > 0$

(4) $\Delta U = 0, q < 0, w > 0$

Q. 34. Radius of the first excited state of helium ion is given as:

$a_0 \rightarrow$ radius of first stationary state of hydrogen atom.

(1) $r = \frac{a_0}{2}$ **(2)** $r = \frac{a_0}{4}$

(3) $r = 4a_0$ **(4)** $r = 2a_0$

Q. 35. Given below are two statements:

Statement I: $CH_3–O–CH_2–Cl$ will undergo S_N1 reaction though it is a primary halide.

StatementII : $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}} - CH_2 - Cl$

will not undergo S_N^2 reaction very easily though it is a primary halide.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is incorrect but Statement II is correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Both Statement I and Statement II are correct.

Q. 36. Given below are two statements:

Statement I: One mole of propyne reacts with excess of sodium to liberate half a mole of H_2 gas.

Statement II: 4g of propyne reacts with $NaNH_2$ to liberate NH_3 gas, which occupies 224 mL at STP.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is incorrect but Statement II is correct.

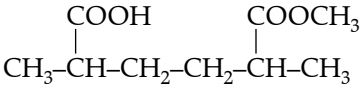
(4) Both Statement I and Statement II are correct.

Q. 37. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of CO_2 is converted into CO on addition of graphite. If total pressure at equilibrium is 0.8 atm, then K_p is:

(1) 0.18 atm **(2)** 1.8 atm

(3) 0.3 atm **(4)** 3 atm

Q. 38. The IUPAC name of the following compound is:



(1) 2-carboxy-5-methoxycarbonylhexane.

(2) methyl-6-carboxy-2,5-dimethylhexanoate.

(3) methyl-5-carboxy-2-methylhexanoate.

(4) 6-methoxycarbonyl-2,5-dimethylhexanoic acid.

Q. 39. Which of the following electrolyte can be used to obtain $H_2S_2O_8$ by the process of electrolysis?

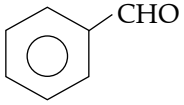
(1) Dilute solution of sodium sulphate

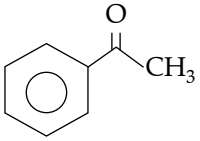
(2) Dilute solution of sulphuric acid

(3) Concentrated solution of sulphuric acid

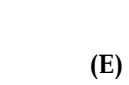
(4) Acidified dilute solution of sodium sulphate

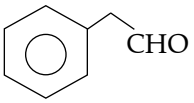
Q. 40. The compounds which give positive Fehling's test are:

(A) 

(B) 

(C) $HOCH_2–CO–(CHOH)_3–CH_2–OH$

(D) 

(E) 

Choose the correct answer from the options given below:

(1) (A), (C) and (D) only

(2) (A), (D) and (E) only

(3) (C), (D) and (E) only

(4) (A), (B) and (C) only

Q. 41. In which of the following complexes the CFSE, Δ_0 will be equal to 0?

(1) $[Fe(NH_3)_6]Br_2$ **(2)** $[Fe(en)_3]Cl_3$

(3) $K_4[Fe(CN)_6]$ **(4)** $K_3[Fe(SCN)_6]$

Q. 42. Arrange the following solutions in order of their increasing boiling points:

(i) 10^{-4} M NaCl (ii) 10^{-4} M Urea

(iii) 10^{-3} M NaCl (iv) 10^{-2} M NaCl

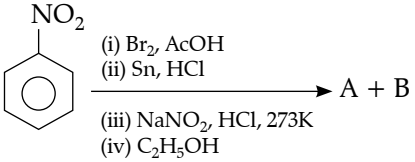
(1) (ii) < (i) < (iii) < (iv)

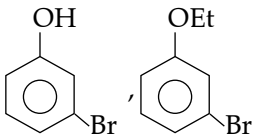
(2) (iii) < (i) < (ii) < (iv)

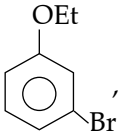
(3) (i) < (ii) < (iii) < (iv)

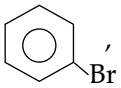
(4) (iv) < (iii) < (i) < (ii)

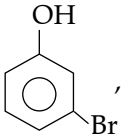
Q. 43. The products formed in the following reaction sequence are:



(1) 

(2)  , CH_3–COOH

(3)  , CH_3–CHO

(4)  , CH_3–CHO

Q. 44. From the magnetic behaviour of $[NiCl_4]^{2-}$ (paramagnetic) and $[Ni(CO)_4]$ (diamagnetic), choose the correct geometry and oxidation state.

(1) $[NiCl_4]^{2-}$: Ni(II), square planar
$[Ni(CO)_4]$: Ni(0), square planar

(2) $[NiCl_4]^{2-}$: Ni(II), tetrahedral
$[Ni(CO)_4]$: Ni(0), tetrahedral

(3) $[NiCl_4]^{2-}$: Ni(II), tetrahedral
$[Ni(CO)_4]$: Ni(II), square planar

(4) $[NiCl_4]^{2-}$: Ni(0), tetrahedral
$[Ni(CO)_4]$: Ni(0), square planar

Q. 45. The incorrect statements regarding geometrical isomerism are:

(A) Propene shows geometrical isomerism.

(B) Trans isomer has identical atoms/groups on the opposite sides of the double bond.

(C) Cis-but-2-ene has higher dipole moment than trans-but-2-ene.

(D) 2-methylbut-2-ene shows two geometrical isomers.

(E) Trans isomer has lower melting point than cis isomer.

Choose the correct answer from the options given below:

(1) (A), (D) and (E) only

(2) (C), (D) and (E) only

(3) (B) and (C) only

(4) (A) and (E) only

Section B

Q. 46. Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K. This entire amount of CO_2 gas was later passed through an aqueous solution of $Ca(OH)_2$. The excess unreacted $Ca(OH)_2$ was later neutralised with 0.1 M of 40 mL HCl. If the volume of the sealed container of CO_2 was x, then x is _____ cm^3 (nearest integer).

(Given: The entire amount of CO_2(g) reacted with exactly half the initial amount of $Ca(OH)_2$ present in the aqueous solution.)

Q. 47. In Carius method for estimation of halogens, 180 mg of an organic compound produced 143.5 mg of AgCl. The percentage composition of chlorine in the compound is _____ %.

(Given: Molar mass in g mol^{-1} of Ag: 108, Cl = 35.5)

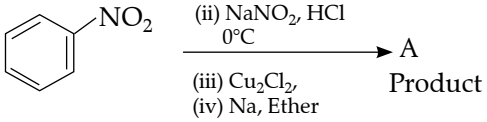
Q. 48. The number of molecules/ions that show linear geometry among the following is _____ .

SO_2, $BeCl_2$, CO_2, N_3^-, NO_2, F_2O, XeF_2, NO_2^+, I_3^-, O_3

Q. 49. A $\rightarrow$ B

The molecule A changes into its isomeric form B by following a first order kinetics at a temperature of 1000 K. If the energy barrier with respect to reactant energy for such isomeric transformation is 191.48 kJ mol^{-1} and the frequency factor is 10^{20}, the time required for 50%, molecules of A to become B is _____ picoseconds (nearest integer). ($R = 8.314\ J\ K^{-1}\ mol^{-1}$)

Q. 50. Consider the following sequence of reactions:



Molar mass of the product formed (A) is _____ g mol^{-1}.

Mathematics

Section A

Q. 51. The number of non-empty equivalence relations on the set {1, 2, 3} is:

(1) 6 **(2)** 7 **(3)** 5 **(4)** 4

Q. 52. Let f: R→R be a twice differentiable function such that $f(x + y) = f(x) f(y)$ for all $x, y \in$ R. If $f'(0) = 4a$ and f satisfies $f''(x) - 3a f'(x) - f(x) = 0, a > 0$, then the area of the region R = $\{(x, y) \mid 0 \le y \le f(ax), 0 \le x \le 2\}$ is:

(1) $e^2 - 1$ **(2)** $e^4 + 1$

(3) $e^4 - 1$ **(4)** $e^2 + 1$

Q. 53. Let the triangle PQR be the image of the triangle with vertices (1, 3), (3, 1) and (2, 4) in the line $x + 2y = 2$. If the centroid of ΔPQR is the point (α, β), then 15(α – β) is equal to:

(1) 24 **(2)** 19 **(3)** 21 **(4)** 22

Q. 54. Let z_1, z_2 and z_3 be three complex numbers on the circle $|z| = 1$ with $\arg(z_1) = \frac{-\pi}{4}$, $\arg(z_2) = 0$ and $\arg(z_3) = \frac{\pi}{4}$.

If $|z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1|^2 = \alpha + \beta\sqrt{2}$, $\alpha, \beta \in Z$, then the value of $\alpha^2 + \beta^2$ is:

(1) 24 **(2)** 41 **(3)** 31 **(4)** 29

Q. 55. Using the principal values of the inverse trigonometric functions the sum of the maximum and the minimum values of $16((\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2)$ is:

(1) $24\pi^2$ **(2)** $18\pi^2$ **(3)** $31\pi^2$ **(4)** $22\pi^2$

Q. 56. A coin is tossed three times. Let X denote the number of times a tail follows a head. If μ and σ^2 denote the mean and variance of X, then the value of $64(\mu + \sigma^2)$ is:

(1) 51 **(2)** 48 **(3)** 32 **(4)** 64

Q. 57. Let $a_1, a_2, a_3, \ldots$ be a GP of increasing positive terms. If $a_1a_5 = 28$ and $a_2 + a_4 = 29$, then a_6 is equal to:

(1) 628 **(2)** 526 **(3)** 784 **(4)** 812

Q. 58. Let $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ be two lines. Then which of the following points lies on the line of the shortest distance between L_1 and L_2?

(1) $\left(-\frac{5}{3}, -7, 1\right)$ **(2)** $\left(2, 3, \frac{1}{3}\right)$

(3) $\left(\frac{8}{3}, -1, \frac{1}{3}\right)$ **(4)** $\left(\frac{14}{3}, -3, \frac{22}{3}\right)$

Q. 59. The product of all solutions of the equation $e^{5(\log_e x)^2 + 3} = x^8$, $x > 0$, is:

(1) $e^{8/5}$ **(2)** $e^{6/5}$ **(3)** e^2 **(4)** e

Q. 60. If $\sum_{r=1}^{n} T_r = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$, then $\lim_{n\to\infty} \sum_{r=1}^{n} \left(\frac{1}{T_r}\right)$ is equal to:

(1) 1 **(2)** 0 **(3)** $\frac{2}{3}$ **(4)** $\frac{1}{3}$

Q. 61. From all the English alphabets, five letters are chosen and are arranged in alphabetical order. The total number of ways, in which the middle letter is 'M', is:

(1) 14,950 **(2)** 6084 **(3)** 4356 **(4)** 5148

Q. 62. Let $x = x(y)$ be the solution of the differential equation $y^2dx + \left(x - \frac{1}{y}\right)dy = 0$. If $x(1) = 1$, then $x\left(\frac{1}{2}\right)$ is:

(1) $\frac{1}{2} + e$ **(2)** $\frac{3}{2} + e$

(3) $3 - e$ **(4)** $3 + e$

Q. 63. Let the parabola $y = x^2 + px - 3$, meet the coordinate axes at the points P, Q and R. If the circle C with centre at (–1, –1) passes through the points P, Q and R, then the area of ΔPQR is:

(1) 4 **(2)** 6 **(3)** 7 **(4)** 5

Q. 64. A circle C of radius 2 lies in the second quadrant and touches both the coordinate axes. Let r be the radius of a circle that has centre at the point (2, 5) and intersects the circle C at exactly two points. If the set of all possible values of r is the interval (α, β), then 3β – 2α is equal to:

(1) 15 **(2)** 14 **(3)** 12 **(4)** 10

Q. 65. Let for $f(x) = 7\tan^8x + 7\tan^6x - 3\tan^4x - 3\tan^2x$, $I_1 = \int_0^{\pi/4} f(x)dx$ and $I_2 = \int_0^{\pi/4} xf(x)dx$. Then $7I_1 + 12I_2$ is equal to:

(1) 2π **(2)** π **(3)** 1 **(4)** 2

Q. 66. Let $f(x)$ be a real differentiable function such that $f(0) = 1$ and $f(x + y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in R$. Then $\sum_{n=1}^{100} \log_e f(n)$ is equal to:

(1) 2384 **(2)** 2525 **(3)** 5220 **(4)** 2406

Q. 67. Let $A = \{1, 2, 3, ..., 10\}$ and

$$B = \left\{\frac{m}{n} : m, n \in A, m < n \text{ and } \gcd(m, n) = 1\right\}$$

Then $n(B)$ is equal to:

(1) 31 **(2)** 36 **(3)** 37 **(4)** 29

Q. 68. The area of the region inside the circle $\left(x - 2\sqrt{3}\right)^2 + y^2 = 12$ and outside the parabola $y^2 = 2\sqrt{3}x$ is:

(1) $6\pi - 8$ **(2)** $3\pi - 8$

(3) $6\pi - 16$ **(4)** $3\pi + 8$

Q. 69. Two balls are selected at random one by one without replacement from a bag containing four white and six black balls. If the probability that the first selected ball is black, given that the second selected ball is also black, is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is equal to:

(1) 14 **(2)** 4 **(3)** 11 **(4)** 13

Q. 70. Let the foci of a hyperbola be (1, 14) and (1, –12). If it passes through the point (1, 6), then the length of its latus rectum is:

(1) $\frac{25}{6}$ **(2)** $\frac{24}{5}$ **(3)** $\frac{288}{5}$ **(4)** $\frac{144}{5}$

Section B

Q. 71. Let the function, $f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ a^2 + bx, & x \geq 1 \end{cases}$ be differentiable for all $x \in R$, where $a > 1$, $b \in R$. If the area of the region enclosed by $y = f(x)$ and the line $y = -20$ is $\alpha + \beta\sqrt{3}$, $\alpha, \beta, \in Z$, then the value of $\alpha + \beta$ is _____.

Q. 72. If $\sum_{r=0}^{5} \frac{^{11}C_{2r+1}}{2r+2} = \frac{m}{n}$, $gcd(m, n) = 1$, then $m - n$ is equal to _____.

Q. 73. Let A be a square matrix of order 3 such that $det(A) = -2$ and $det(3adj(-6adj(3A))) = 2^{m+n}.3^{mn}$, $m > n$. Then $4m + 2n$ is equal to _____.

Q. 74. Let $L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0}$ and $L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha}$, $\alpha \in R$, be two lines, which intersect at the point B. If P is the foot of perpendicular from the point A(1, 1, –1) on L_2, then the value of $26\,\alpha(PB)^2$ is ____.

Q. 75. Let $\vec{c}$ be the projection vector of $\vec{b} = \lambda\hat{i} + 4\hat{k}$, $\lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $|\vec{a} + \vec{c}| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____.

■■■